

UTILISING OPEN SOURCE

For Cloud-Agnostic Delivery Of IoT Edge

As the world becomes increasingly interconnected, the demand for efficient and secure communication between devices grows ever more critical. Thin edge computing, with its ability to perform computation at the edge of the network, has emerged as a key enabler of the Internet of Things (IoT). In this article, we explore the fascinating intersection of thin edge and IoT, and examine how this powerful combination is driving innovation and transforming industries



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As we all know, the Internet of Things (IoT) and cloud computing have been revolutionising all kinds of industries and have introduced so many business models as well. They are allowing developers to introduce new ideas.

Today, manufacturing is going through a fundamental transition. Gone are the days where post facto analysis of defects would happen; today it is more about precision manufacturing, more importantly predictive defect analysis. Most of the time the technology that lets you do that is either on the device side, edge side, or the cloud side.

India is now making big moves in smart metering. Also, India is making investments to make changes in our

consumption pattern of energy, such as oil and electricity. All of which is being done to improve the efficiency in consumption. And they are being driven by edge computing semantics.

Need for continuous optimisation

Let us understand the need for continuous optimisation through an analogy. A remote wind farm has to be operated manually. You are often not ready to pay the cost of an operator going there, because it is too far off. But, at the same time, you need to introduce a lot of intelligence to those wind farms.

Such situations require you to be at the pinnacle of the technology and use edge computing along with IoT. Fortunately, there has been a huge logistical shift. Many

service providers have started bringing in new technologies and are providing new services at better cost-efficiency, because of which this revolution has happened. There is a huge opportunity for developers to fix the old problems using IoT computing.

When the challenges are big then the opportunities are big too. In this case also, there are huge challenges that persist. Let us take a look at some of them.

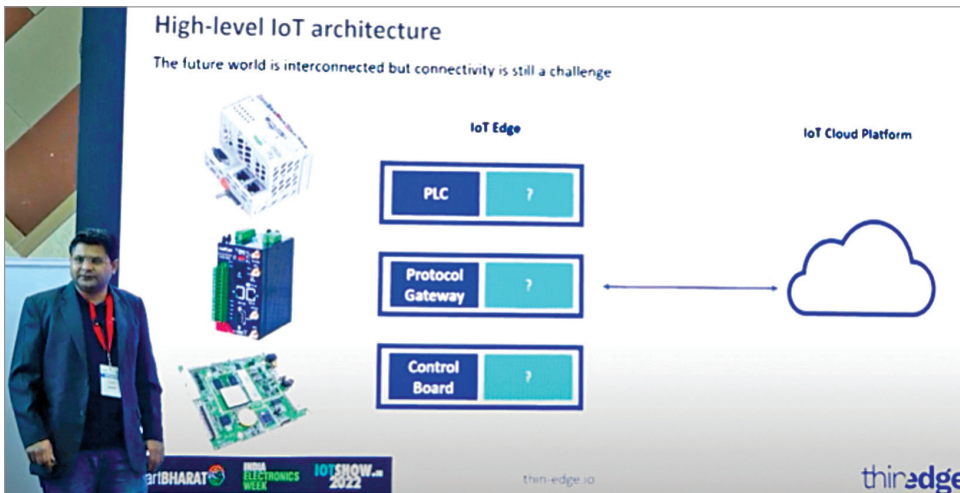


Fig. 1: High-level IoT architecture